Advarra Research Portal API Overview

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Summary

This document provides an overview of the REST-based (Representational State Transfer-based) Advarra Research Portal (ARP) API and its technical infrastructure, as developed and provided by Advarra. Readers of this document are assumed knowledgeable in the areas of web-based technologies.

The API exists to facilitate the exchange of data between OnCore and a public-facing web page used for discovery of ongoing research studies.

Architecture

The REST-based API employs a design based on some of the key principles from the representational state transfer architectural style and utilizes Hypertext Transfer Protocol Secure (HTTPS) as the transport protocol. The API is deployed within OnCore in the same application server instance.

Security

Since the API is designed to support public-facing web pages, no authentication is required.

All endpoints are limited to HTTP GET operations. No part of the API allows creating or updating data in OnCore.

All API service operation exchanges are encrypted using Secure Socket Layer (SSL), the same technology used to encrypt all web browser-based communication with the OnCore application.

Implementation and Maintenance

Customers are responsible for developing, testing, and maintaining the API client. New versions of OnCore may require code changes to the API client. Time and resources need to be allocated when performing OnCore upgrades to ensure workflows designed around the API continue to work as expected.

Getting Started

The API uses the OpenAPI specification to define the data model and interface endpoints that are available in the REST-based API. The use of OpenAPI allows for the use of various code generation tools including OpenAPI Generator and Swagger Codegen. Both are open-source projects that allow generation of an API client from an OpenAPI specification.

https://github.com/OpenAPITools/openapi-generator/blob/master/README.md

https://swagger.io/tools/swagger-codegen/

These code generators can use the REST-based API specification to generate client code in a variety of languages (Java, C#, Python, Ruby, etc.).

Available Resources

The Learning Portal includes a link to a .pdf of the Advarra Research Portal API Specification. Below are some notes regarding how to use the REST API Specification and general information applicable across the various resources.

API Specification Information

The records returned from a search are not ordered.

Search parameters for GET operations and reference list lookups are case insensitive unless noted otherwise in the specification.

Using the API From a Browser

Unauthenticated GET operations are easy to execute from a standard web browser just using the address bar.

The steps below show how to do a basic GET of a protocol from ARP in a web browser.

For this example, we will do a search for an existing protocol.

1. In the browser address bar, enter the URL for which to perform a GET on a protocol. The URL will be a combination of the main OnCore URL that is used to login: for example,
   https://myoncore.forteresearchapps.com
   followed by the BasePath as defined in the Advarra Research Portal API Specification: **/arp/api/**
   followed by the URL for searching for a protocol, as defined in the Advarra Research Portal API Specification: **/protocol**
   In this case, the URL would be:
   https://myoncore.forteresearchapps.com/arp/api/protocol
   This endpoint requires a query parameter to specify which protocol to retrieve. Add a **?** to the end of the URL and add a query parameter where the Key is protocolNo and the Value is a known protocol number in the OnCore environment: for example, **?protocolNo=ONC0116**.
   The full URL would then be:
   https://myoncore.forteresearchapps.com/arp/api/protocol?protocolNo=ONC0116
2. Press Enter.
3. The response will contain the protocol information.

https://myoncore.forteresearchapps.com/arp/api/protocol?protocolNo=ONC0116

Pretty-print ☑


```
{
  "protocolNo": "ONC0116",
  "status": "Open",
  "title": "A Phase III Randomized Trial of Adjuvant &#8364;Chemotherapy With or Without Bevacizumab for Patients With Comp
adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient mont
massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a
elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, eni
Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam
ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec
leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue veli
ut, mollis sed, nonummy id, metus. Nullam accumsan lorem in dui. Cras ultricies mi eu turpis hendrerit fringilla. Vestibulu
lacinia. Nam pretium turpis et arcu. Duis arcu tortor, suscipit eget, imperdiet nec, imperdiet iaculis, ipsum. Sed aliquam
ullamcorper ipsum rutrum nunc. Nunc nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl
mollis hendrerit risus. Phasellus nec sem in justo pellentesque facilisis. Etiam imperdiet imperdiet orci. Nunc nec neque.
vitae, posuere imperdiet, leo. Maecenas malesuada. Praesent congue erat at massa. Sed cursus turpis vitae tortor. Donec pos
ultrices posuere cubilia Curae; Sed aliquam, nisi quis porttitor congue, elit erat euismod orci, ac placerat dolor lectus q
dolor. ",
  "objective": null,
  "secondaryProtocolNos": [
    "GSK-16121-005",
    "NCI-2009-00509"
  ],
  "nctId": "NCT00324805",
  "sponsorTypes": [
    "Industry",
    "National"
```


Making CORS Exceptions

Default configurations of OnCore will respond with security warnings if trying to include content in an external site. To make a CORS exception, an Apache configuration can be added for the external site. For example, on the OnCore application server:

/opt/forte/apache2.4/conf.d/70-cors.conf

This file would have Apache directives for allowing the website to query SIP data. For example:

```
<Location "/arp/api">
   Header always set Access-Control-Allow-Origin "https://website.mydomain.edu"
</Location>
```

For OnCore applications hosted on Advarra Managed Infrastructure, contact your Advarra Product Support Specialist to request a CORS exception for your website.